

Microsoft Fabric High Availability Business Continuity Disaster Recovery BCDR Accelerator



Session Agenda:

- ❑ Reliability in Azure 101
- ❑ What is BCDR
- ❑ High Availability vs Disaster Recovery
- ❑ Shared Responsibility Model
- ❑ Reliability Features in Microsoft Fabric
- ❑ BCDR Accelerator overview and demo

What is BCDR?



BCDR stands for Business Continuity and Disaster Recovery. It refers to a set of processes and strategies that organizations use to ensure they can continue operating during and after a disaster.

BCDR helps organizations mitigate the impact of various disruptions, whether they are natural disasters, cyberattacks, or other unexpected events.

High Availability (HA) vs Disaster Recovery (DR)



High Availability (HA) focuses on minimizing downtime and ensuring that systems remain operational during minor failures or maintenance.

Disaster Recovery (DR), on the other hand, is about restoring systems and data after a major failure or disaster.

[High availability and disaster recovery overview | Microsoft Learn](#)

[Recommendations for designing for redundancy - Microsoft Azure Well-Architected Framework | Microsoft Learn](#)

Microsoft Global Cloud Infrastructure

30+

Azure
Geographies

60+

Azure
regions

300+

Datacenters
worldwide

275K+ Miles
of fiber

190+

Network
PoPs

- Available region
- ◆ Availability Zones available
- ◆ Announced Availability Zones
- ⊗ Announced region with Availability Zones
- Edge Site

WAN Links
Microsoft Confidential

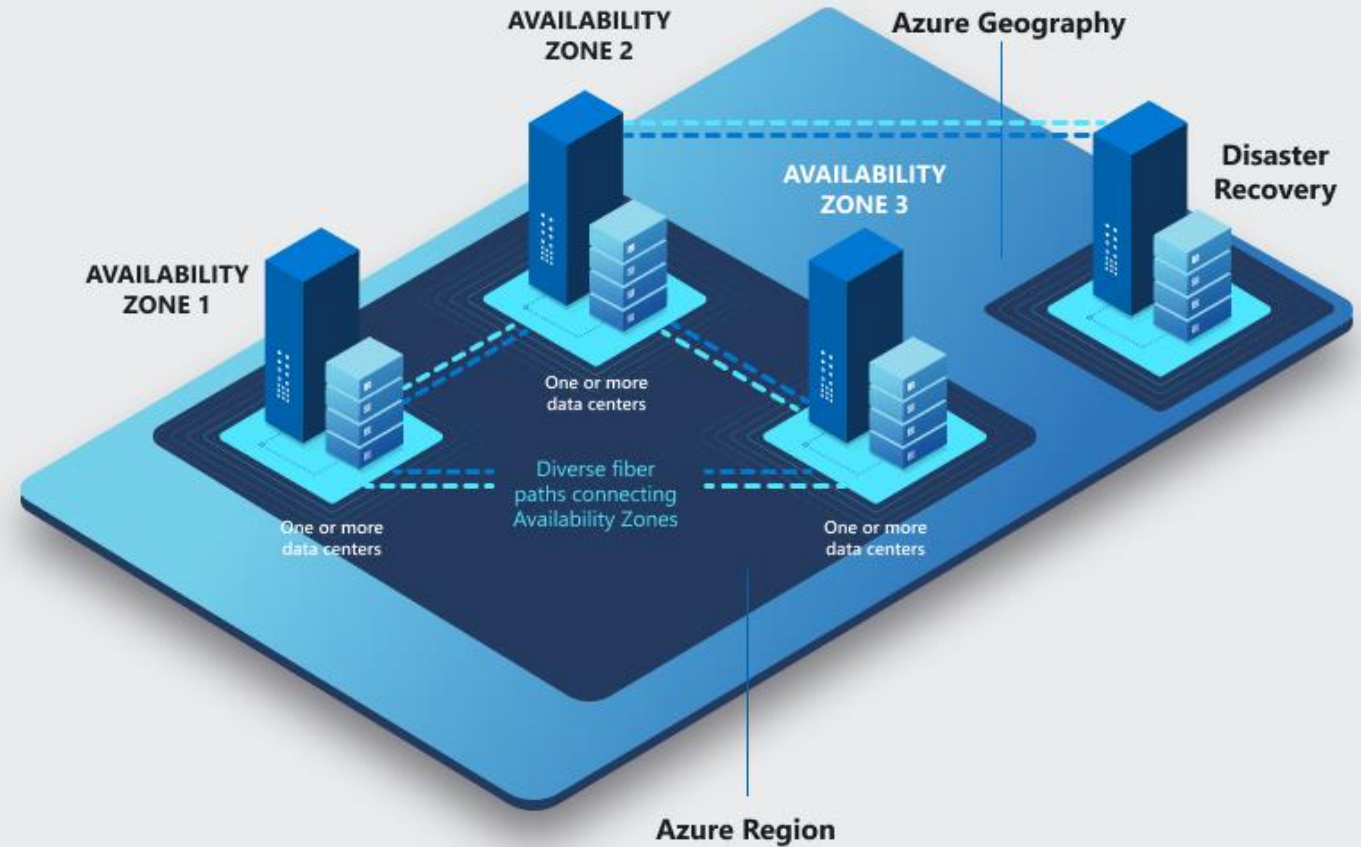
Azure Geographies

Regions

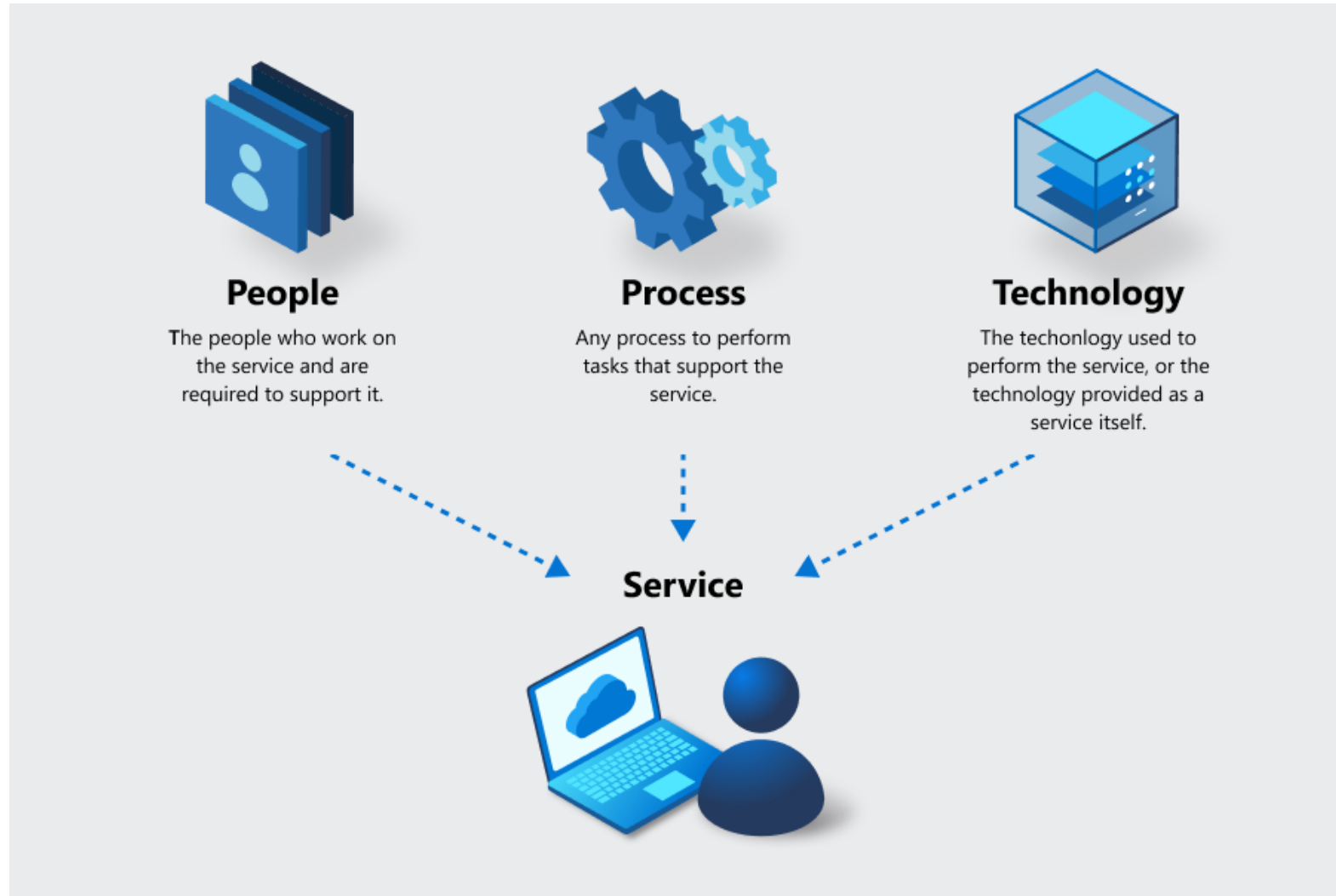
Region-pairs

Availability Zones

Data centers



Business continuity is a shared responsibility



Shared responsibility model

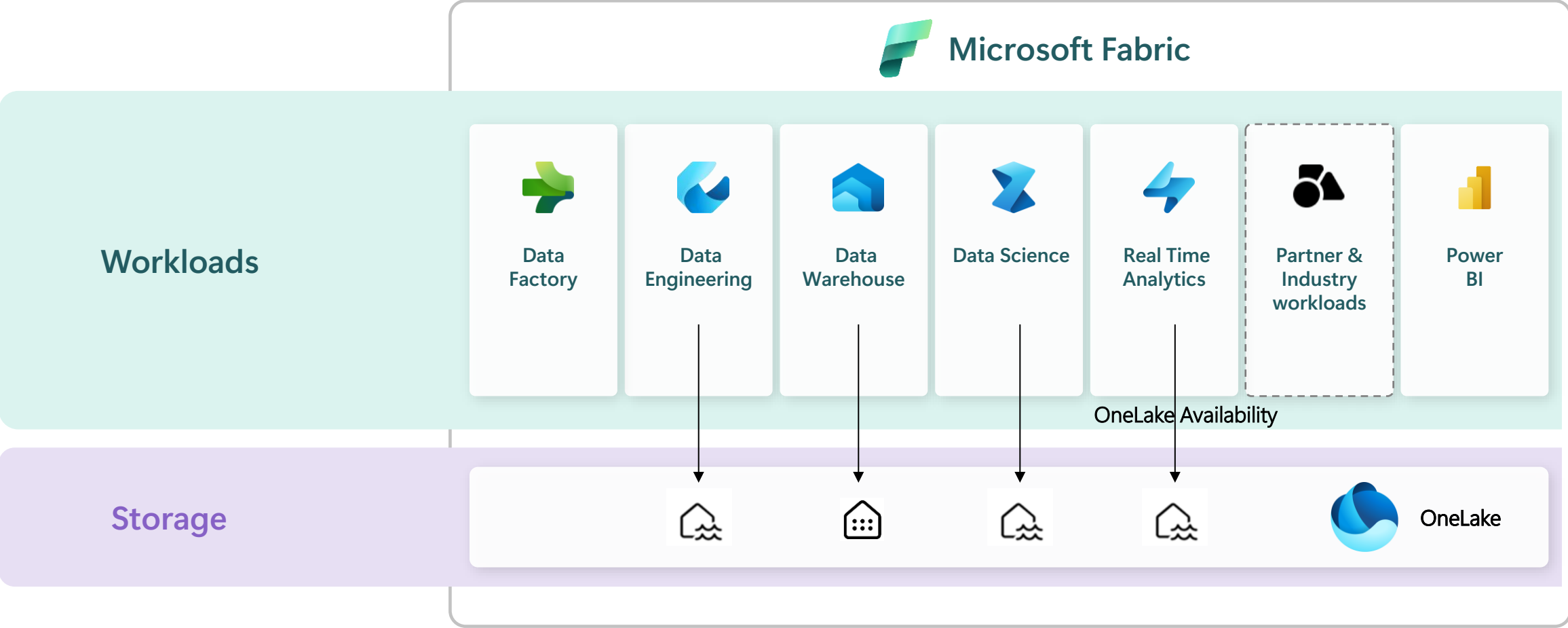
Responsibility	SaaS	PaaS	IaaS	On-prem	
Information and data	Customer	Customer	Customer	Customer	RESPONSIBILITY ALWAYS RETAINED BY CUSTOMER
Devices (Mobile and PCs)	Customer	Customer	Customer	Customer	
Accounts and identities	Customer	Customer	Customer	Customer	
Identity and directory infrastructure	Microsoft	Microsoft	Customer	Customer	RESPONSIBILITY VARIES BY SERVICE TYPE
Applications	Microsoft	Microsoft	Customer	Customer	
Network controls	Microsoft	Microsoft	Customer	Customer	
Operating system	Microsoft	Microsoft	Customer	Customer	
Physical hosts	Microsoft	Microsoft	Microsoft	Customer	RESPONSIBILITY TRANSFERS TO CLOUD PROVIDER
Physical network	Microsoft	Microsoft	Microsoft	Customer	
Physical datacenter	Microsoft	Microsoft	Microsoft	Customer	

Microsoft

Customer

Business Continuity features in Microsoft Fabric

Microsoft Fabric Overview



Features which support reliability and recovery

- Power BI already includes read-only disaster recovery capabilities*
- [Availability Zone support](#) (in supported regions per workload)
- Workspace [Git integration](#) (metadata only)
- Datawarehouse [Restore in-place](#)
- Time travel in [Lakehouse](#) and [Warehouse](#)
- OneLake [ZRS/LRS](#)
- OneLake [Soft Delete](#)
- Disaster Recovery – [OneLake cross region replication](#)

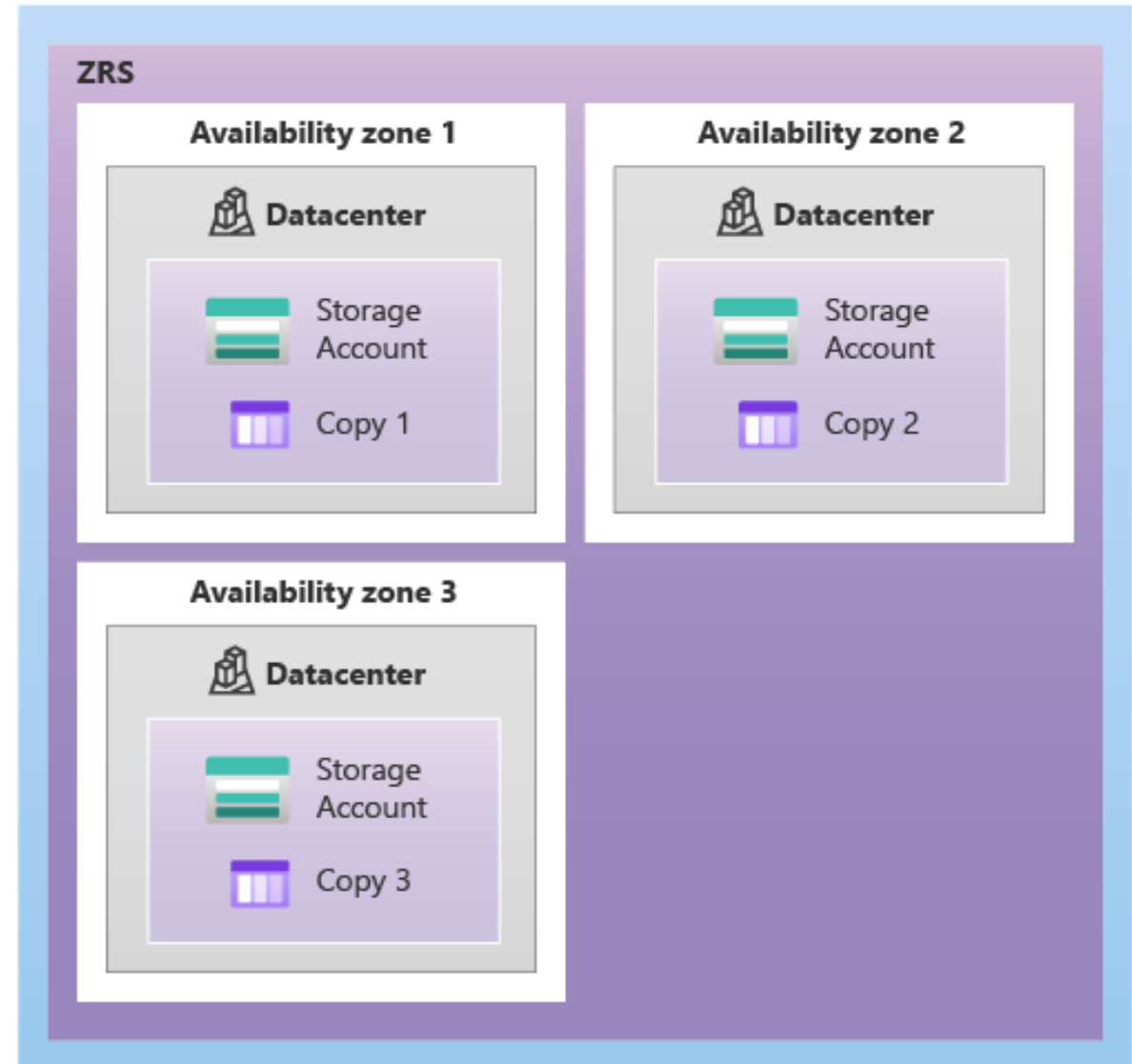
OneLake – Data Redundancy

OneLake utilizes Zone Redundant Storage (**ZRS**) within [supported regions](#) providing at least **99.9999999999%** (12 9's) durability of objects over a given year. [\[1\]](#) [\[2\]](#)

With both LRS and ZRS storage, data is resilient to transient hardware failures within a data center.

With ZRS, your data has fault tolerance to entire data center failures.

Primary region



Fabric – Availability Zone Support

During a zone-wide outage, no action is required during zone recovery.

Fabric capabilities in [supported regions](#) self-heal and rebalance automatically to take advantage of the healthy zone.

Running Spark Jobs may fail if the master node is in the failed zone. In such a case, the jobs will need to be resubmitted.

Americas	Power BI	Datamarts	Data Warehouses	Real-Time Analytics	Data Factory (pipelines)	Data Engineering
Brazil South	✓		✓	✓	✓	
Canada Central	✓			✓	✓	
Central US	✓					✓
East US	✓	✓	✓	✓	✓	✓
East US 2	✓	✓	✓	✓	✓	✓
South Central US	✓				✓	
West US 2	✓	✓			✓	
West US 3	✓			✓	✓	

Enabling DR at Fabric capacity Level

Admin portal

Tenant settings New

Usage metrics

Users

Premium Per User

Audit logs

Domains New

Workloads

Capacity settings

Refresh summary

Embed Codes

Organizational visuals

Azure connections

Workspaces

Custom branding

Protection metrics

Fabric identities

Featured content

Details Delegated tenant settings

Disaster Recovery

Turn on this setting to enable disaster recovery for this capacity. With disaster recovery on, data sources used in Fabric items are replicated to secondary regions, providing access to your organization's critical data in case of a service interruption. Fabric items include, but are not limited to, lakehouses, warehouses, notebooks, and Apache Spark job definitions. Fabric disaster recovery may not be available in your region.

Fabric disaster recovery is a paid feature and will appear on your statement as data replication metered charges. [Learn more](#)

i This setting does not apply to or impact BCDR capabilities previously released with Power BI.

– **Unable to change selection** This setting was turned on 0 days ago and can be changed only once in 30 days. You can change the setting again in 30 days.

☒ On

Capacity usage report

OneLake which logically spans the world

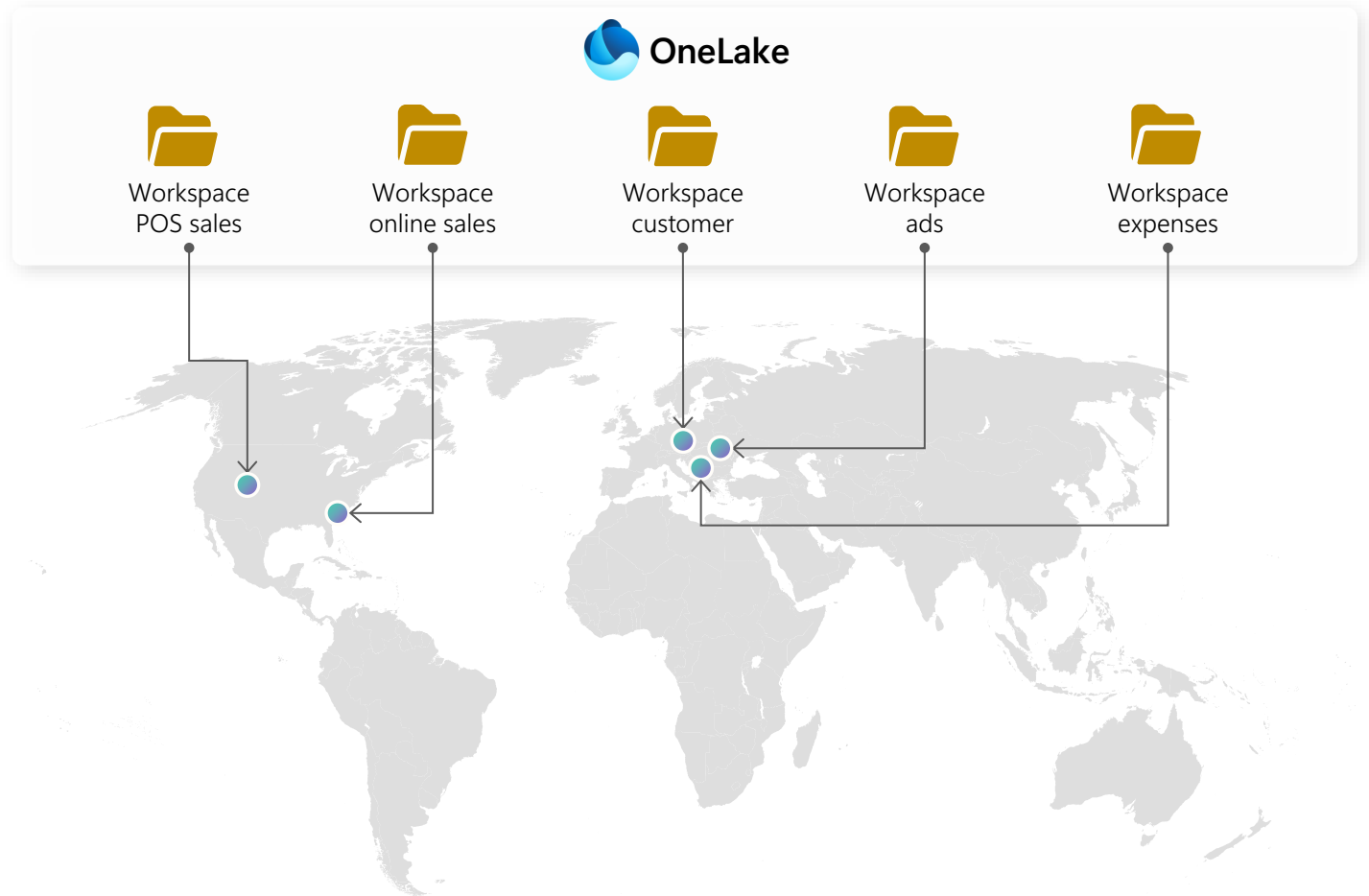
To achieve data residency requirements, workspaces can reside in different regions around the world while still being part of the same data lake.

Data can reside in different regions without the overhead of managing different storage resources and without creating data silos.

OneLake provisions storage resources for each workspace to meet demand for scale (capacity, throughput and IOPS).

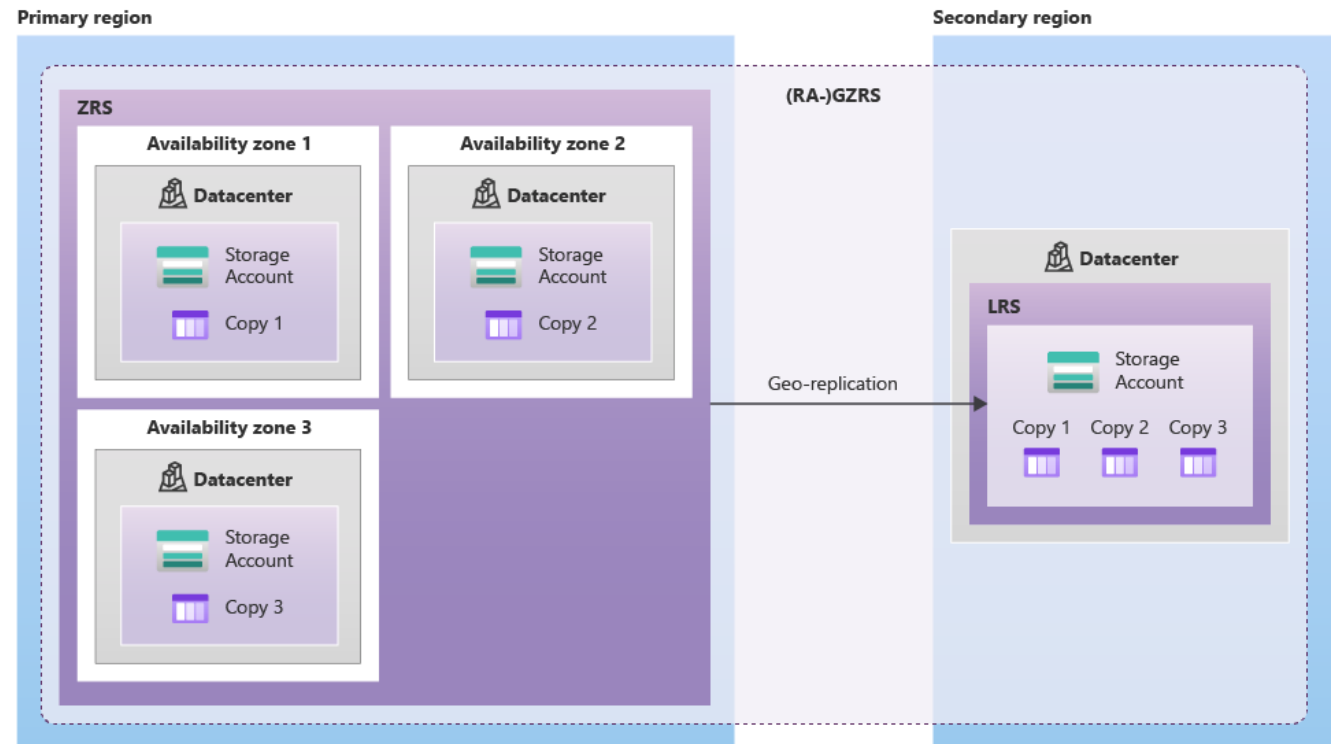
Underlying physical storage is virtualized away.

All storage is zone redundant by default with an option for Geo redundancy.



OneLake cross region replication (DR enabled)

OneLake with cross region replication (DR enabled at capacity level) utilizes Geo-Zone Redundant Storage (**GZRS**) within [supported regions](#) providing at least 99.999999999999999% (16 9's) durability of objects over a given year where supported [\[1\]](#) [\[2\]](#)



OneLake cross region replication (DR enabled)

- OneLake data (includes Lakehouse and warehouse) is asynchronously replicated and stored in the paired geographic region, making it geo-redundant (GRS).
- Increased [OneLake consumption](#)- write operations consume additional capacity units due to replication
- Increased [OneLake storage](#) charges
- Possible network egress charges (tbd)
- When a major disaster renders the primary region unrecoverable, Microsoft Fabric initiates a regional failover.
- When failover completes, data can be accessed in the secondary region through OneLake's APIs via the global endpoint e.g.:

`https://onelake.dfs.fabric.microsoft.com/<workspaceGUID>/<itemGUID>/<path>/<fileName>`
`abfs[s]://<capacity.workspace>@onelake.dfs.fabric.microsoft.com/<item>.<itemtype>/<Tables>/<fileName>`

- After a failover, the new primary data center will have local redundancy only. Failing back to the original primary may (once it has been restored) involve executing the DR plan in reverse.

Disaster/Primary Outage: State of Fabric

Fabric portal

- Portal **read** operations supported
- Write operations paused

Power BI

- **Read** operations supported
- Write operations not supported

Lakehouses/Warehouses

- **Inaccessible** but files can only be accessed using OneLake APIs or tools. Items need to be recovered.

Notebooks

- **Inaccessible** and needs to be recovered

Dataflows Gen2/Pipelines/Eventstreams

- **Inaccessible** and needs to be recovered

Disaster/Primary Outage: State of Fabric cont.

Spark Job Definition

- **Inaccessible**
- Code files can be accessed via OneLake APIs

ML Model/Experiment

- **Inaccessible**
- Code content and metadata (such as run metrics and configurations) won't be saved after the disaster.

KQL Database/Queryset

- **Inaccessible**
- Additional prerequisite steps required to protect data in the same.

Building reliable analytics is a shared responsibility

Customer analytics, workloads and data

Customer's **analytics, workloads** and **data** including backups & version control

Customer's responsibility
Reliability *in* the cloud

Utilising reliability features

Optional capabilities that **customers can enable and utilise as needed**—git integration and disaster recovery, and backup

Resilient foundation

Core Azure capabilities **built into the platform**—how the foundation is designed, operated, and monitored to ensure availability

Our responsibility
Reliability *of* the cloud

Customer's responsibility: Prepare and Plan for disasters



Activate Disaster Recovery Capacity Settings

Decide on DR region – usually region pair

Regularly review and set the disaster recovery capacity settings

Ensure settings meet protection and recovery needs



Create Data Backups

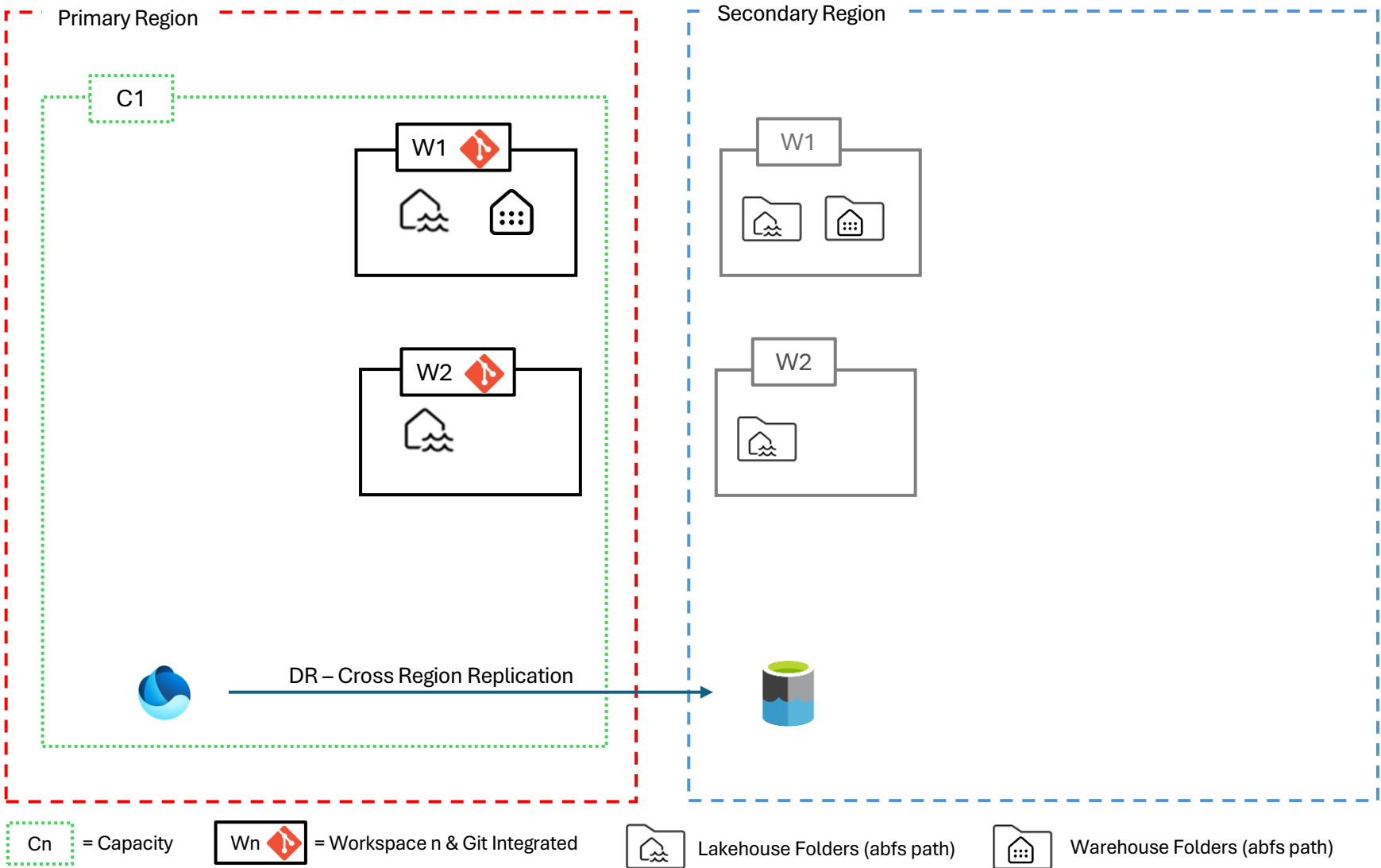
Copy critical data stored outside of OneLake to another region aligned with your disaster recovery plan



Build & test the recovery process

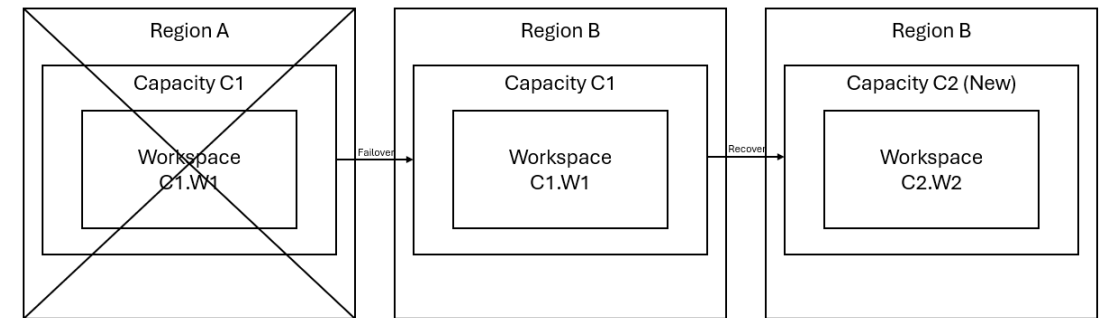
Develop and regularly review (DR drills) the recovery process

Before: DR scenario with DR setting enabled

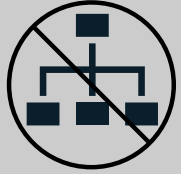


Experience-specific disaster recovery guidance

1. Create a new Fabric capacity C2 in a new region.
2. Create a new W2 workspace in C2, including its corresponding items with same names as in C1.W1.
3. Copy data from the disrupted C1.W1 to C2.W2.
4. Follow the dedicated instructions for each component to restore items to their full function.



Challenges with the disaster recovery guidance



Doesn't scale: Customers usually have multiple capacities and hundreds of workspaces



Requires a manual approach: Current guidance involves clickops – doesn't scale, takes too long, error prone



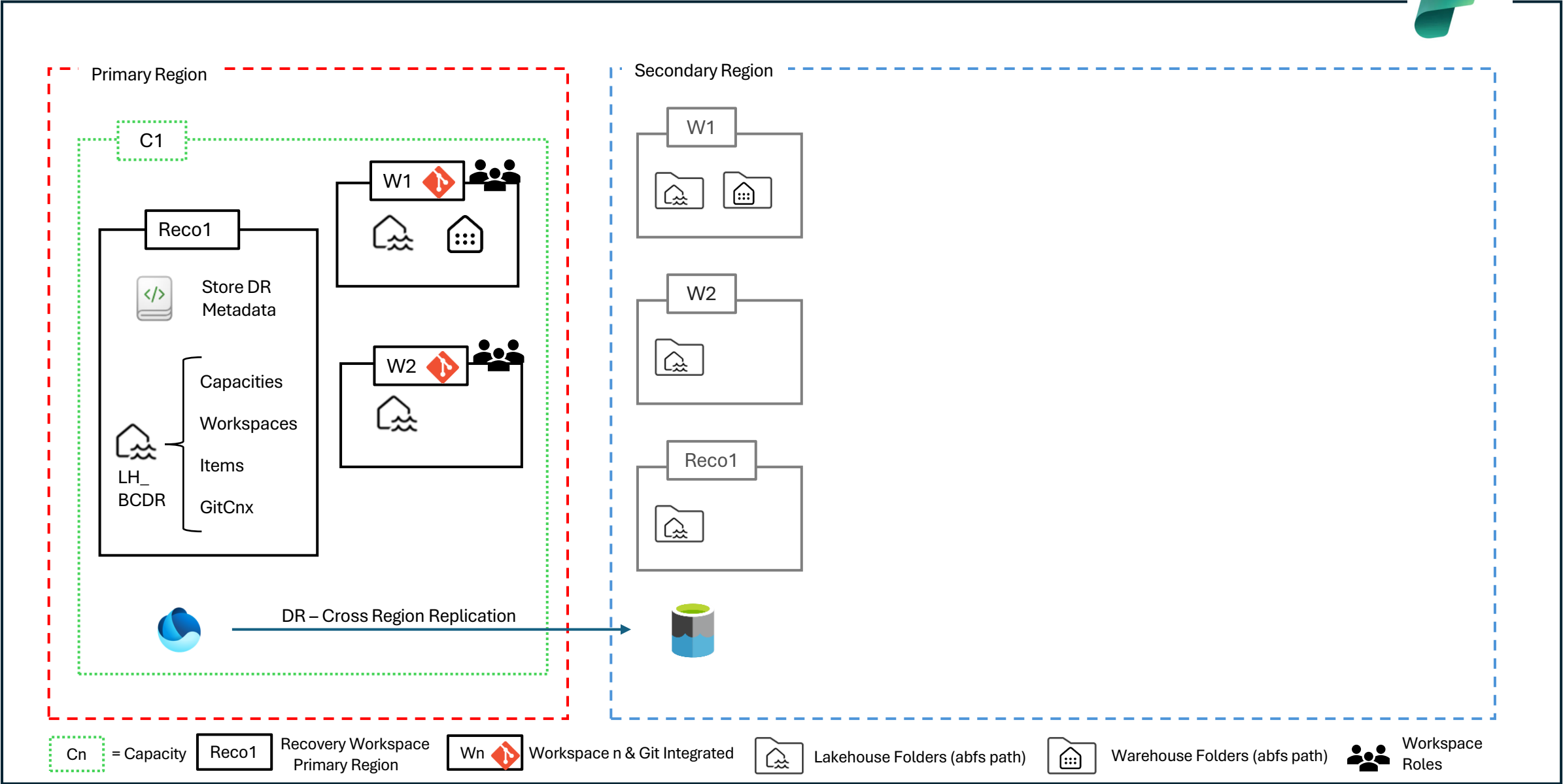
Sign-off: Regulated industries require a robust, regularly tested and efficient DR strategy

Introducing the Microsoft BCDR Accelerator...

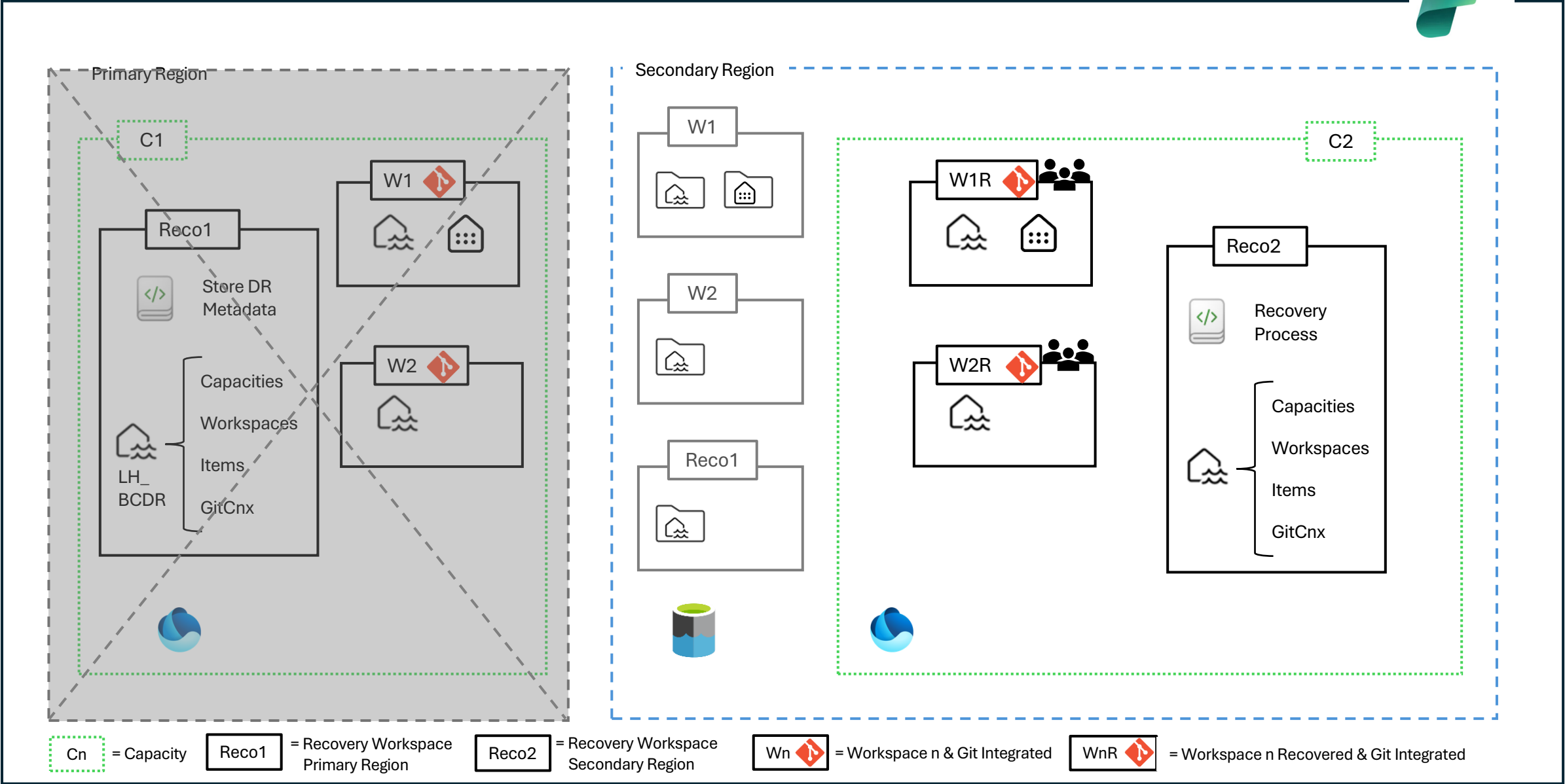
A community-based accelerator which provides an automated process using Fabric notebooks to recover Git supported Fabric items and associated data in another region.



Before: DR scenario with DR setting enabled + recovery metadata



After fail-over and recovery – Fabric items and data recovered. Workspace roles applied



BCDR Accelerator Demonstration



Useful Links

- Fabric BCDR Accelerator VBD – aka.ms/bcdracceleratorvbd
- Fabric BCDR Blog - aka.ms/FabricBCDRPlanningBlog
- <https://learn.microsoft.com/en-us/fabric/onelake/onelake-disaster-recovery>
- <https://learn.microsoft.com/en-us/fabric/security/experience-specific-guidance>
- <https://azure.microsoft.com/en-gb/pricing/details/microsoft-fabric/>



Questions?